

7th Set of Problems

1. Consider a desired path, $p_d(s)$, which is parameterized along it's length s , such as the one illustrated in Figure 1(a). It is desired to move along the path from lengths s_o to s_f , with null velocity at the beginning and the end of the trajectory, in an interval of time t_f . Consider a single cubic polynomial to design the timing law for the trajectory's length, $s = s(t)$.

- Deduce the equations for coefficients a_i , as functions of s_o , s_f and t_f .
- Considering the velocity constraint, $|\dot{s}(t)| < \dot{s}_{max}$, where $\dot{s}_{max}$ is constant and positive. Determine the minimum value for t_f , as a function of s_o , s_f and $\dot{s}_{max}$.
- Considering the acceleration constraint, $|\ddot{s}(t)| < \ddot{s}_{max}$, where $\ddot{s}_{max}$ is constant and positive. Determine the minimum value for t_f , as a function of s_o , s_f and $\ddot{s}_{max}$.

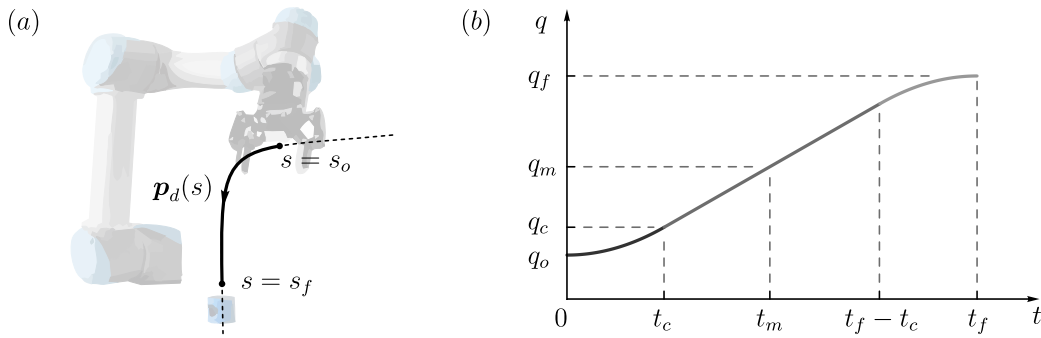


Figure 1: (a) Robot's desired path, $p_d(s)$, parameterized along it's length. (b) Linear trajectory with parabolic boundaries.

2. It is desired to move the joint of a robot from q_o to q_f , with null velocity at the beginning and the end of the trajectory, in an interval of time t_f . Consider the linear trajectory with parabolic boundaries in Figure 1(b), where the values for q_o , q_f and t_f are known, while t_c is left for parameterization.

- Deduce the equations for the linear and parabolic sections, as functions of q_o , q_f , t_c and t_f .
- Considering the velocity and acceleration constraints, $|\dot{q}(t)| < \dot{q}_{max}$, $|\ddot{q}(t)| < \ddot{q}_{max}$, where $\dot{q}_{max}$ and $\ddot{q}_{max}$ are constant and positive. Determine the minimum and maximum values for t_c , as a function of q_o , q_f , t_f , $\dot{q}_{max}$ and $\ddot{q}_{max}$.

3. Consider the Cartesian space trajectory illustrated in Figure 2(a).

- (a) Parameterize the piecewise trajectory as a function of its length.
- (b) Design a cubic temporal law to span the trajectory in a period of time t_f .
- (c) Describe a general placement of the trajectory in space.

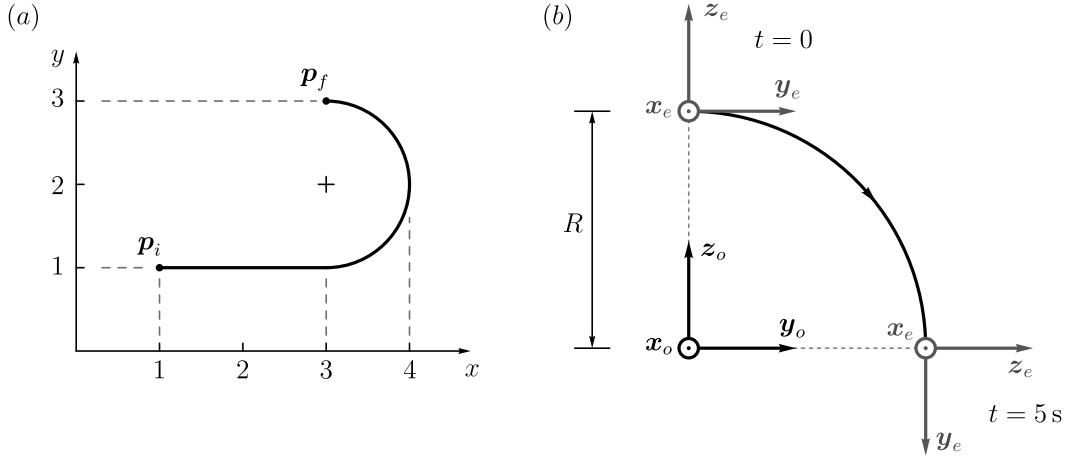


Figure 2: (a) Piecewise trajectory. (b) Arc trajectory with tangent frame.

4. For a given manipulator, it is desired to perform a trajectory in an arc, keeping y_e tangent to the arc, according to Figure 2(b). The timing law should be cubic, starting and ending with zero velocity, with total time 5 s. Deduce the expressions for the position and orientation trajectory, using for the orientation the vector-angle representation.

Solutions: 1.(a) $s(t) = s_o + 3(s_f - s_o)(t/t_f)^2 - 2(s_f - s_o)(t/t_f)^3$; (b) $t_f > \frac{3|s_f - s_o|}{2\dot{s}_{max}}$; (c) $t_f > \sqrt{\frac{6|s_f - s_o|}{\ddot{s}_{max}}}$. 2.(a) $q(t) = \{q_o + a t^2, 0 \leq t < t_c; q_m + b(t - t_m), t_c \leq t < t_f - t_c; q_f - a(t - t_f)^2, t_f - t_c \leq t < t_f\}$ with $q_m = q_o + \frac{q_f - q_o}{2}$, $t_m = \frac{t_f}{2}$, $a = \frac{q_m - q_o}{t_c(2t_m - t_c)}$ and $b = 2a t_c$; (b) $\max\left(0, \frac{t_f}{2} - \sqrt{\frac{t_f^2}{4} - \frac{|q_f - q_o|}{\ddot{q}_{max}}}\right) < t_c < \min\left(\frac{t_f}{2}, t_f - \frac{|q_f - q_o|}{\ddot{q}_{max}}\right)$. 3.(a) $p(s) = \left\{ \begin{pmatrix} 1 + s \\ 1 \end{pmatrix}, 0 \leq s \leq 2; \begin{pmatrix} 3 + \sin(s - 2) \\ 2 - \cos(s - 2) \end{pmatrix}, 2 < s \leq 2 + \pi \right\}$. 4. $p_d(s) = \begin{pmatrix} 0 \\ R \sin(\frac{s}{R}) \\ R \cos(\frac{s}{R}) \end{pmatrix}$ and $R_d(s) = R_x\left(-\frac{s}{R}\right) \forall 0 \leq s \leq \frac{\pi R}{2}$ with $s(t) = \frac{\pi R}{250}(15 - 2t)t^2$.